

Dr. Tom DE GEUS



PROFILE

Dedicated and innovative **scientist** with a passion for tackling **complex problems** through creative **interdisciplinary** solutions. Disposing of extensive experience in **mechanics, physics, materials, and mathematics & programming**. Excelling at identifying the core of problems, enabling fundamental solutions. **Project management**: budget & prioritization, resulting in efficient & on-time progress. **Supervisor**: fostering a productive and collaborative environment, through confidence, patience, and respect.

CONTACT DETAILS

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PERSONAL INFORMATION

Year of birth: **1985**

Citizenship: **The Netherlands**

Residency: **Switzerland** (C permit)

Civil status: **Married**

Languages:

Dutch (native), **English** (C2),
French (C2), **German** (C2/C1)

SKILLS

- Problem evaluation & solving
- Project management
- Leadership
- Communication
- Coaching & teaching
- Python, C++, git, LaTeX, ...
- Big data, statistics, FEM

EXPERIENCE

- **PATENT EXPERT & EXAMINER** **2024–pres.**
Swiss Federal Institute of Intellectual Property (Switzerland)
Examined patents in various subjects in French and German (written and oral communication with inventors and their lawyers).
- **SENIOR RESEARCHER PHYSICS** **2019–2024**
EPFL (Switzerland) – FNSF Ambizione
Acquired and managed funding (567 kCHF), independently designed and executed research resulting in top journal publications and conference presentations, independently supervised bachelor and master students, co-supervised PhD student, designed and applied high-performance computations, reviewed for top journals, datachampion.
Subjects: tribology, fracture, earthquakes, glasses, complex systems.
- **POSTDOCTORAL RESEARCHER PHYSICS** **2016–2019**
EPFL (Switzerland) – NWO Rubicon
Acquired funding (160 k€), published in top journals, participated in education and student supervision, designed and applied computations.
- **POSTDOCTORAL RESEARCHER VALORIZATION** **2016**
Mzi & TATA Steel (The Netherlands)
Acquired funding (10 k€), developed numerical tool for industrial usage.
- **RESEARCH INTERNSHIP** **2010**
Harvard University (United States)
Designed and analyzed 3D meta-material, developed numerical tools.

EDUCATION

- **PHD MECHANICAL ENGINEERING (CUM LAUDE)** **2012–2016**
Eindhoven University of Technology (The Netherlands)
Specialisation: materials (mechanics & fracture) and numerical solvers.
Martinus van Marum award for best PhD thesis (12.5 k€), various awards.
- **MSc-BSc MECHANICAL ENGINEERING (GREAT APPRECIATION) 2004–2012**
Eindhoven University of Technology (The Netherlands)
Specialisation: mechanics of materials. Minor in entrepreneurship, winner entrepreneurship award (1 k€).

PROBLEM SOLVING EXPERIENCE

- **ANALYTICAL THINKING**: big data from high-performance computing.
- **CRITICAL THINKING**: proposed **new theory friction**.
- **CREATIVITY**: proposed novel simple models by abstraction.
- **ADAPTABILITY**: changed scientific fields, driver heavy equipment.

PROJECT MANAGEMENT EXPERIENCE

- **PRIORITIZE**: finds balance ad-hoc and permanent solutions.
- **BUDGET**: managed research project and sport club budgets.
- **ACHIEVING RESULTS**: accomplished goals **Ambizione grant**.
- **ACCOUNTABILITY**: group leader mountaineering & rowing outings.
- **PERSISTENCE & RESILIENCE**: 29 publications, Swiss rowing champion.

LEADERSHIP EXPERIENCE

- **INSPIRE**: challenging student projects, **world's longest rowing race**.
- **MOTIVATE**: co-supervisor PhD-student, team member.
- **COMMUNICATE**: Interviews: Horizons, RTS (TV). Open-source coding.
- **GUIDE**: 32 bachelor and master projects, tutor high school.
- **INCLUSION**: buddy autistic student, trainer diverse rowers.
- **DELEGATE**: shift leader catering team football stadium PSV.